Seeing Through Clouds: Using Radar to Rebuild Cloud-Covered Sentinel-2 Satellite Images

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Abstract—Satellite photos of farmland are often useless because clouds cover the ground, and this happens most during the monsoon, exactly when farmers and planners need the data. Radar satellites can see through clouds, but radar images do not look like photos. We built a deep learning model, called dieWolke, that takes a cloudy Sentinel-2 image together with a Sentinel-1 radar image of the same place and predicts what the ground looks like without clouds, in all 13 spectral bands that Sentinel-2 records. On top of a standard design we tested four ideas: processing related bands in separate branches, checking a vegetation index (NDVI) during training, letting the model say how confident it is in each pixel, and finding the clouds first in a separate step. We trained on a small dataset (2,434 image patches) and tested on 1,500 patches from places the model had never seen. Our model comes within 0.5–0.7 dB of two published models that were trained on far more data, and beats a classical machine-learning method by more than 2 dB. Finding the clouds first is clearly the right choice (1.1 dB better than skipping that step), and the confidence map is genuinely useful: ignoring the least confident half of the pixels cuts vegetation-index errors by 26 %. The vegetation-index check helps a little, and splitting the bands into branches did not help with this amount of data.

Index Terms—cloud removal, satellite imagery, radar, Sentinel-1, Sentinel-2, deep learning, agriculture

I. Introduction

The European Sentinel-2 satellites photograph the whole Earth every few days. Unlike a normal camera, which records three colours, they record 13 bands: the visible colours plus several invisible ones in the near- and short-wave infrared. Those invisible bands are what make the images so useful for farming, because healthy plants reflect strongly in the near infrared and wet soil looks different from dry soil in the short-wave infrared. A simple number computed from the red and near-infrared bands, the NDVI (Normalised Difference Vegetation Index), is the standard way to measure how green and healthy crops are.

The problem is clouds. In India's Kharif season (July to October), the main monsoon growing season, most Sentinel-2 images of farmland are partly or completely cloudy. The data is missing exactly when it matters most.

The Sentinel-1 satellites carry a radar instead of a camera. Radar sends out its own microwave signal, which passes through clouds, so Sentinel-1 always sees the ground. Radar images, however, show surface texture and moisture rather than colour, so they cannot simply replace the photo.

The idea behind this project, and behind earlier work such as DSen2-CR [2], GLF-CR [3] and UnCRtainTS [4], is to give a neural network both images: the cloudy photo shows what the clear parts look like, and the radar gives clues about what is hidden under the clouds. The network learns to fill in the gaps.

What we add. Combining radar and optical images is not new, and neither is the type of model we use (a so-called diffusion bridge [5]). What we contribute are four additions aimed at agricultural use, each tested carefully on its own:

- Band groups (C1): the visible, the plant-sensitive (red-edge and near infrared) and the moisture-sensitive (short-wave infrared) bands are handled by separate branches of the network, because they behave differently.
- Vegetation check (C2): during training, the model is also penalised when the NDVI of its output is wrong, not only when the pixels are wrong.
- Confidence map (C3): the model outputs, for every pixel, how sure it is of its own answer, so users know which parts to trust.
- Clouds first (C5): a small separate network first finds the clouds and their shadows, and only then does the main network fill them in.

A fifth part of the project (C4), a test set built from real Indian Kharif fields, is in progress and not covered here.

How we test. All versions of our model share the same design, training time and data, so when we switch one idea off, any change in the results comes from that idea alone. We also ran the published UnCRtainTS and GLF-CR models, using their authors' own trained weights, through exactly the same test code. Ideas that did not work are reported as such.

II. Related Work

Radar plus optical. DSen2-CR [2] was one of the first deep networks to combine Sentinel-1 radar with cloudy Sentinel-2 images, together with the SEN12MS-CR dataset of matching radar, cloudy and clear images

from around the world [1]. GLF-CR [3] improved how the two kinds of image are combined, and UnCRtainTS [4] added an estimate of how uncertain each prediction is.

Diffusion models. Diffusion models are the family behind modern AI image generators. A diffusion bridge is a variant that learns to turn one image into another, here a cloudy image into a clear one [6]. DB-CR [5] applied this to cloud removal with radar, and CloudBreaker [7] produces four optical bands from radar alone.

Classical methods. Gupta and Verma [8] filled cloudy Sentinel-2 pixels in India with tree-based machine learning before classifying crops. Our XGBoost comparison [9] follows the same spirit, but it is our own implementation, not theirs.

III. How the Model Works

A. Inputs and outputs

For each location the model receives three things: the cloudy Sentinel-2 image (13 bands), a Sentinel-1 radar image (two channels, called VV and VH), and a cloud mask, a black-and-white map of which pixels are covered by cloud or cloud shadow. It outputs the 13-band cloud-free image and, optionally, a confidence map. All images are 256×256 pixels, or 2.56 km by 2.56 km on the ground. Pixel values are scaled to the range 0 to 1.

B. The bridge between cloudy and clear

Picture a slider with the cloudy image at one end and the clear image at the other. Positions in between are a blend of the two plus a little random noise. During training, the network is shown points along this slider and has to work out the clear end. Formally, the point at position t (0 = clear, 1 = cloudy) is

$$x_t = (1-t)\,x_{\text{clear}} + t\,x_{\text{cloudy}} + \sigma\sqrt{t(1-t)}\,\epsilon, \qquad (1)$$

where ϵ is random noise and $\sigma = 0.5$ sets how much noise is added. At $t = 1$ the point is exactly the cloudy image, with no noise. When the model is used, it starts there and jumps to the clear end in a single step, which makes it fast. All results in this paper use this one-step mode. During training, a quarter of the examples are taken at exactly $t = 1$ so that the model gets plenty of practice at this one-step jump.

C. The network

The main network is a U-Net, a standard design for image tasks that first shrinks the image to understand the overall scene and then expands it back to full size to produce detailed output. It has four levels of detail. A separate radar branch turns the radar image into features at every level, and attention layers at the coarser levels let the optical part of the network look up the relevant radar information. Pixels that are not cloudy are copied straight from the input, so the network only has to fill in what is hidden. Training minimises the average difference between the predicted and the true clear image (an L1 loss), plus the extra terms described below.

D. C1: Band groups

In the full model the input layer, every attention layer and the output layer are split into three branches: visible (bands B01–B04), red-edge and near infrared (B05–B08, B8A, B09) and short-wave infrared (B10–B12). Each branch has its own weights and its own view of the radar features, and the branches share a short summary with each other. The comparison version replaces the branches with a single shared layer of about the same size.

E. C2: Vegetation check

The loss gets an extra term: the difference between the NDVI of the predicted image and the NDVI of the true image, with weight 0.1. NDVI is a ratio, and on very dark pixels (water, shadow, missing data) its value jumps wildly; in our first runs this made training break down. We therefore leave out pixels where the red and near-infrared bands are both very dark.

F. C3: Confidence map

For each pixel and band, the model also predicts a spread b: a small b means "I am sure", a large b means "I might be far off". It learns this by being rewarded when large errors come with a large spread and small errors with a small spread [10]. This extra task is kept separate from the main prediction, so switching it on does not change what the main image is trained on. After training, the spreads are rescaled slightly so that they match the real errors on held-out data (calibration [11]).

G. C5: Finding the clouds first

On real cloudy images nobody knows the exact cloud outline, so we estimate it with a standard cloud detector, s2cloudless [12], plus a simple rule for shadows [13]. To get images where the right answer is known exactly, we also create synthetic cloudy images: a real cloud is cut out of one image and pasted, with its shadow, onto a clear image of another place. The cloud-finding network (a small U-Net) is trained on these synthetic images and finds the cloud-and-shadow area with an overlap (IoU) of 0.964 with the truth on validation data. For comparison, an end-to-end version gets no cloud mask at all and has to figure out the clouds by itself while reconstructing.

IV. Experiments

A. Data

We use SEN12MS-CR [1], a public dataset of matching radar, cloudy and clear images. To keep the first experiments small and fast, we use only its spring part. From the training areas we took a random sample of 2,500 patches; after removing pairs whose "clear" image was itself more than 5 % cloudy, 2,434 remained for training. We use 300 patches for validation (choosing the best checkpoint) and 1,500 patches from five test areas that the model never sees during training. Because we use only spring data, our numbers should not be compared directly with papers that use all four seasons, such as DB-CR [5].

B. Training

Each model was trained on free NVIDIA T4 graphics cards on Kaggle, with batches of 8 random 128×128 crops (flipped and rotated for variety), the AdamW optimiser with a learning rate of 2×10^{-4} that slowly decreases, and a smoothed average of the weights. The full model and the comparison model trained for 40,000 steps; the experiments that switch one idea off trained for 20,000 steps each, against a 20,000-step copy of the full model. Each version was trained once.

C. What we compare

- Ours, full: with C1, C2 and C3 switched on.
- Ours, matched: the identical model with C1, C2 and C3 switched off. This is the fair test of whether our additions help.
- UnCRtainTS and GLF-CR: published models, run with their authors' weights. They were trained on all four seasons of SEN12MS-CR, far more data than ours. As a sanity check, UnCRtainTS scores 28.83 dB on our spring test set, close to the 28.90 dB its authors report on the full test set [4], which tells us we are feeding it data correctly.
- XGBoost: a classical machine-learning method that, for each image, learns from the clear pixels how radar relates to each band, then fills in the cloudy pixels.
- Cloudy input: doing nothing, as a floor.

D. How we score

- PSNR (peak signal-to-noise ratio, in dB): how close the output is to the true image; higher is better, and in this field a gain of 1 dB is considered substantial.
- SSIM (0 to 1): how well shapes and textures match; higher is better.
- SAM (degrees): how well the "spectral signature" of each pixel, its pattern across the 13 bands, matches; lower is better.
- NDVI error (RMSE): how far off the vegetation index is; lower is better.

All are computed on all 13 bands for each test image and then averaged. "Cloud" versions use only the pixels that were under cloud. When we say one model is better than another, we give a 95 % confidence interval in square brackets, computed by resampling the test images; if the interval includes zero, the difference could be chance.

V. Results

A. Main comparison

Table I shows the results on real clouds. Every deep model improves the image by about 10 dB over doing nothing and beats XGBoost by more than 2 dB. Our full model is slightly behind our matched model (0.20 dB [0.14, 0.27]). The published models are ahead of ours by 0.46–0.71 dB, which is not surprising given that they learned from many times more data. GLF-CR has clearly the most accurate vegetation index.

TABLE I
Real clouds, 1,500 test patches. Higher PSNR and SSIM are better; lower SAM and NDVI error are better. Our models learned from 2,434 spring patches; the published models from all of SEN12MS-CR.

Method	PSNR	SSIM	SAM	NDVI error
Cloudy input (do nothing)	18.62	0.676	13.82	0.269
XGBoost (classical)	25.98	0.803	9.54	0.167
Ours, full	28.12	0.866	8.40	0.153
Ours, matched	28.32	0.870	8.32	0.152
UnCRtainTS (published)	28.83	0.882	8.00	0.145
GLF-CR (published)	28.59	0.887	7.79	0.129

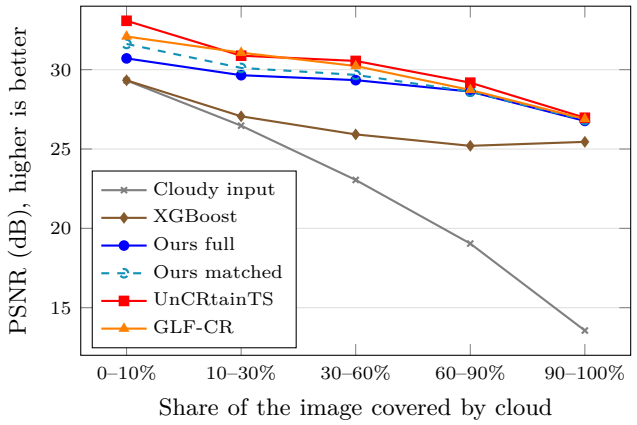


Fig. 1. Quality versus cloud cover on real clouds (139, 148, 196, 299 and 718 test patches per group). Under near-total cloud the deep models are almost equal; the published models lead on partly cloudy images.

B. Where the difference comes from

Fig. 1 splits the test images by how cloudy they are, and this tells an interesting story. When an image is almost completely covered (90 % cloud or more, which is almost half the test set), the model has nothing to go on except the radar, and all four deep models score within 0.2 dB of each other. Our full model is level with our matched model and with GLF-CR there, and only 0.20 dB behind UnCRtainTS.

The published models pull ahead on partly cloudy images, by 0.8–1.2 dB. To do well on those, a model has to combine the visible pixels with the radar, and cope with the fact that the cloudy and the clear photo were taken on different days. That is the kind of skill that comes from seeing many examples, so more training data should help us most here.

The ranking also changes from place to place. In one test area both of our models beat UnCRtainTS, and in the cloudiest area our full model beats both our matched model and GLF-CR. With only five test areas, the overall ranking is less certain than the confidence intervals suggest.

TABLE II
Synthetic clouds (real clouds pasted onto clear images, so the truth is known exactly).

Method	PSNR	SSIM	SAM	NDVI error
Cloudy input	19.51	0.689	8.81	0.226
Ours, full	29.08	0.870	7.49	0.148
Ours, matched	28.88	0.876	7.62	0.151

TABLE III
Switching off one idea at a time (real clouds, 20,000 training steps each). The last column is the change in PSNR compared with the full model, with its confidence interval.

Version	PSNR	NDVI error	Change in PSNR
Full (all ideas on)	27.84	0.152	—
Without band groups	28.20	0.153	+0.36 [0.29, 0.44]
Without vegetation check	28.00	0.157	+0.16 [0.11, 0.20]
Without confidence map	27.99	0.149	+0.15 [0.09, 0.21]

C. Synthetic clouds

With pasted clouds we know the exact truth underneath and the exact cloud outline (Table II). Here the picture flips: our full model is 0.20 dB [0.12, 0.28] ahead of the matched model, with better spectral and vegetation accuracy, although the matched model keeps a slightly better SSIM. So whether our additions help depends on the type of cloud, and neither version is better across the board.

D. Testing each idea on its own

Table III switches off one idea at a time. Surprisingly, switching off any of them makes the overall image slightly better: by 0.36 dB without band groups, 0.16 dB without the vegetation check and 0.15 dB without the confidence map. The vegetation check does do its own job: without it, the NDVI error goes up by about 3.5 % (0.0053 [0.0038, 0.0067]). The band groups, on the other hand, do not even improve the bands they were designed for; the red-edge, near-infrared and short-wave infrared errors are the same or lower without them.

E. Finding the clouds first

This is our clearest result (Table IV). Without a cloud mask, the model is 1.1 dB worse overall [0.98, 1.20], 1.4 dB worse under the clouds, and has a much lower SSIM (0.78 versus 0.86). It is better on only about a quarter of the images. Just as important, the masks from our own cloud-finding network work as well as the exact masks, so the two-step design costs nothing.

F. The confidence map

The confidence map is useful. Pixels where the model says it is unsure really do have larger errors (rank correlation 0.43 on real clouds, 0.55 on synthetic ones), and after calibration its stated confidence matches reality more closely (calibration error 0.055, down from 0.085).

TABLE IV
Finding clouds first versus not using a mask (synthetic clouds).

Version (20,000 steps)	PSNR	SSIM	PSNR under cloud
With the exact cloud mask	28.75	0.863	27.56
With our cloud finder's mask	28.76	0.864	27.58
No mask at all	27.66	0.784	26.20

TABLE V
Keeping only the pixels the model is most confident about (real clouds).

Share of pixels kept	100 %	90 %	75 %	50 %
NDVI error	0.182	0.171	0.158	0.134

The practical test is in Table V: if we throw away the pixels the model is least sure about, the vegetation index of the remaining pixels gets steadily more accurate, and keeping the most confident half cuts the NDVI error by 26 %. One caution: the calibration learned on real clouds made the synthetic-cloud calibration worse, so it does not carry over between cloud types.

VI. Discussion and Limitations

More data is the obvious next step. Our models learned from 2,434 patches; the published models from the whole dataset. Their advantage is concentrated on partly cloudy images, which is where more examples should help most. We are now retraining on a much larger part of the spring training data (which has 24,378 patches in total) to find out whether the gap is about data or about design.

Band groups did not work, for now. Splitting the bands into branches made results slightly worse and did not help the targeted bands. It may need more data to pay off, or it may simply not be a useful idea for this task; we do not claim it as a contribution based on these results.

Things to keep in mind. Each version was trained only once, so small differences of a few tenths of a dB between our own versions might change with another training run. The test set comes from only five areas. The cloud outlines on real images are estimates. And all results are for spring images only.

VII. Conclusion

A radar-guided model can rebuild cloud-covered Sentinel-2 images in all 13 bands well, even when trained on a small dataset: it comes within 0.5–0.7 dB of published models trained on far more data and is more than 2 dB better than a classical method. Two of our ideas clearly earn their place: finding the clouds first, and giving a confidence map that tells users which pixels to trust, which makes the vegetation index 26 % more accurate when the unsure half is set aside. The vegetation check is a small trade-off, and the band groups did not help at this scale. Next, we will train on more data, repeat the key runs,

and test the model on real Indian Kharif fields with a crop classification task.

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